Indian Institute of Technology Jodhpur
Department of Mathematics
Mathematics II MAL1020
Problem Sheet III

Academic Year 2025-2026 Semester II

1. Find a matrix
$$A = \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix}$$
such that
$$A^2 + \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} = I_3.$$

2. If
$$A = \begin{pmatrix} x & 1 \\ -2 & 1 \end{pmatrix},$$
determine all values of x and y for which $A^2 = A$.

3. Let $A = \begin{pmatrix} 2 & 2 & 1 \\ 2 & 5 & 2 \\ 1 & 2 & 2 \end{pmatrix}$, and let S be the matrix with column vectors
$$\mathbf{s}_1 = \begin{pmatrix} -x \\ 0 \\ x \end{pmatrix},\ \mathbf{s}_2 = \begin{pmatrix} -y \\ y \\ -y \end{pmatrix},\ \mathbf{s}_3 = \begin{pmatrix} z \\ 2z \\ z \end{pmatrix},$$
where x, y, z are (real) constants.
   (a) Show that $AS = \begin{pmatrix} \mathbf{s}_1 & \mathbf{s}_2 & 7\mathbf{s}_3 \end{pmatrix}$.
   (b) Find all the values of x, y, z such that
$$S^T AS = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{pmatrix}.$$

4. Use elementary row operations to reduce the given matrix to row-echelon form:

   (i) $\begin{pmatrix} 2 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 3 & 1 & 5 & 7 \end{pmatrix}$.

(ii) $\begin{pmatrix} 4 & 7 & 4 & 7 \\ 3 & 5 & 3 & 5 \\ 2 & -2 & 2 & -2 \\ 5 & -2 & 5 & -2 \end{pmatrix}$. You can try to further find the reduced row-echelon form for both of them.

5. Reduce the given matrix to reduced row-echelon form:

$$\begin{pmatrix} 1 & -2 & 1 & 3 \\ 3 & -6 & 2 & 7 \\ 4 & -8 & 3 & 10 \end{pmatrix}.$$

6. Determine the solution set to

$$\begin{aligned} 5x_1 - 6x_2 + x_3 &= 4 \\ 2x_1 - 3x_2 + x_3 &= 1 \\ 4x_1 - 3x_2 - x_3 &= 5. \end{aligned}$$

7. Determine the solution set to

$$\begin{aligned} x_1 + x_2 - x_3 + x_4 &= 1 \\ 2x_1 + 3x_2 + x_3 \quad\quad &= 4 \\ 3x_1 + 5x_2 + 3x_3 - x_4 &= 5. \end{aligned}$$

8. Let $X = \mathbb{R}^2$. Also, suppose

$$Y = \{(x, 2x) : x \in \mathbb{R}\} \text{ and } z = \{(x, 3x) : x \in \mathbb{R}\}.$$

Can we write $X = Y \oplus Z$? Justify your answer.